

Inflammation, Infection, and the Immune System

Inflammation is a response of the immune system to infection, injury, or a pathogen. Acute inflammation produces pain, fever, and local heat. Chronic inflammation lasts longer and can damage tissue and organs. The immune system uses antibodies, cells, and receptors.

Bacteria and viruses are common pathogens. Bacterial infection may need a drug that the hospital provides after diagnosis. Viral infection often requires supportive therapy because many viruses have no specific treatment. Vaccines prevent several infections by training immune memory.

The nervous system, the lungs, the liver, and the kidney can all show inflammatory disease. A biopsy and imaging support the diagnosis when symptoms are not specific. Cancer may also create an inflammatory environment around a tumor.

Clinical research describes pathways and mechanisms. A receptor on a cell can start a cascade of proteins. If this process remains active, chronic disease follows. New therapy tries to block the pathway without harming healthy tissue.

Safety remains important. Adverse events during treatment must be reported. The physician and the nurse watch the patient for severe reactions after a dose. This article treats inflammation as a central process in modern medicine.